SHAMANTH S

shamanthsheni@gmail.com| +91 8891429533 | LinkedIn URL | GitHub URL

SUMMARY

AI/ML undergraduate with hands-on experience in **Machine Learning, Deep Learning, NLP, Computer Vision, and AI-powered applications**. Experienced in research-oriented experimentation, developing end-to-end intelligent systems, and deploying automated AI pipelines using cloud and DevOps tools. Currently working on **nuclear pleomorphism analysis in computational pathology** as a research intern. Strong interest in AI research, applied machine learning, and building production-oriented AI systems.

EDUCATION

B.E in Artificial Intelligence and Machine Learning **CGPA – 7.7** **June 2027 ***
Jyothy Institute of Technology (VTU), Bengaluru
Coursework: Mathematics, Machine Learning, Artificial Intelligence, Data Science, Software Engineering

Pre-University (PCMB) **Aggregate: 90%** **March 2023**
Ambika Padavipoorva Vidyalaya, Puttur

TECHNICAL SKILLS

- **Programming Languages:** Python, C, MySQL, MongoDB
- **ML&DL:** Machine Learning, PyTorch, CNNs, Transfer Learning, Attention Mechanisms
- **NLP & AI:** Sentence-BERT, BERTopic, Embeddings
- **Computer Vision:** Image Processing, Computational Pathology
- **Devops:** AWS, Docker, Github, Linux

EXPERIENCE

Research Intern – Computational Engineering Lab, CIIRC ***May 2026 – Present***

- Contributing to research on **nuclear pleomorphism and automated Nottingham grading** in histopathological images.
- Experimented with **SkDenseNet, U-Net, CellViT, and StarDist** for nucleus detection, segmentation, and classification.
- Analyzing **tumor nuclear morphology** for quantitative pleomorphism assessment.
- Reviewing research papers to understand **state-of-the-art architectures and methodologies** in computational pathology.

PROJECTS

MSA-CSRNet — Multi-Scale Crowd Counting

• Reimplemented CSRNet in **PyTorch** with VGG16 and dilated convolutions for density-based crowd counting on ShanghaiTech Part A.

• Enhanced the architecture with **multi-scale feature fusion** and **SE attention**, reducing **MAE from 73.78 to 62.81** (14.9% improvement) and achieving **r = 0.957.**

Strategic Research Miner — AI Research Intelligence Platform

• Engineered an AI pipeline analyzing **80,000+ research papers** using Sentence-BERT and BERTopic to uncover emerging research trends and potential gaps.

• Integrated **LSTM forecasting (MAE ≈ 0.077)** with Strategic Scoring, hypothesis generation, and real-time validation against **2024–2025 arXiv data**.

BharathBrief — AI Problem Intelligence Platform

• Engineered an automated pipeline using **Google Gemini** to transform signals from **250+ news and public sources** into structured, actionable problem briefs.

• Built and deployed a **Next.js + Supabase** platform processing **1,000+ extracted problems**, with **20 curated** till date and **GitHub Actions** automating daily execution.

CERTIFICATES

- **Oracle AI Foundations –** Oracle (Oct 2025) – [Link]
- **AWS Academy Foundations-**AWS (Oct 2025) – [Link]
- **C for Everyone - University of California, Santa Cruz** (Mar 2025) – [Link]